Cardio Regulations 2: Study Notes

Course: Personal Training Certification

Topic: Cardio Regulations, part 2 of 2

About these notes: Short revision version of the Cardio Regulations 2 lecture. For the full explanation of every concept, example and class question, see the Detailed Notes. Sources: the lecture recording, the course slides ("Cardio Regulations"), the board photos and the E-book chapter on cardiovascular training techniques.

Big idea: In cardio, **intensity is the heart rate during the activity, expressed as a percentage of the person's own MHR**. The level (50 to 60%, 60 to 70%, 70 to 85%, above 85%) is chosen from six factors and is converted into a **target heart rate range** with the age-predicted method or, better, the **HRR (Karvonen) method**. The client follows the range with a device, or by RPE and the talk test when numbers are risky. The main phase of a session can be continuous, LIIT/MIIT, HIIT or Fartlek.

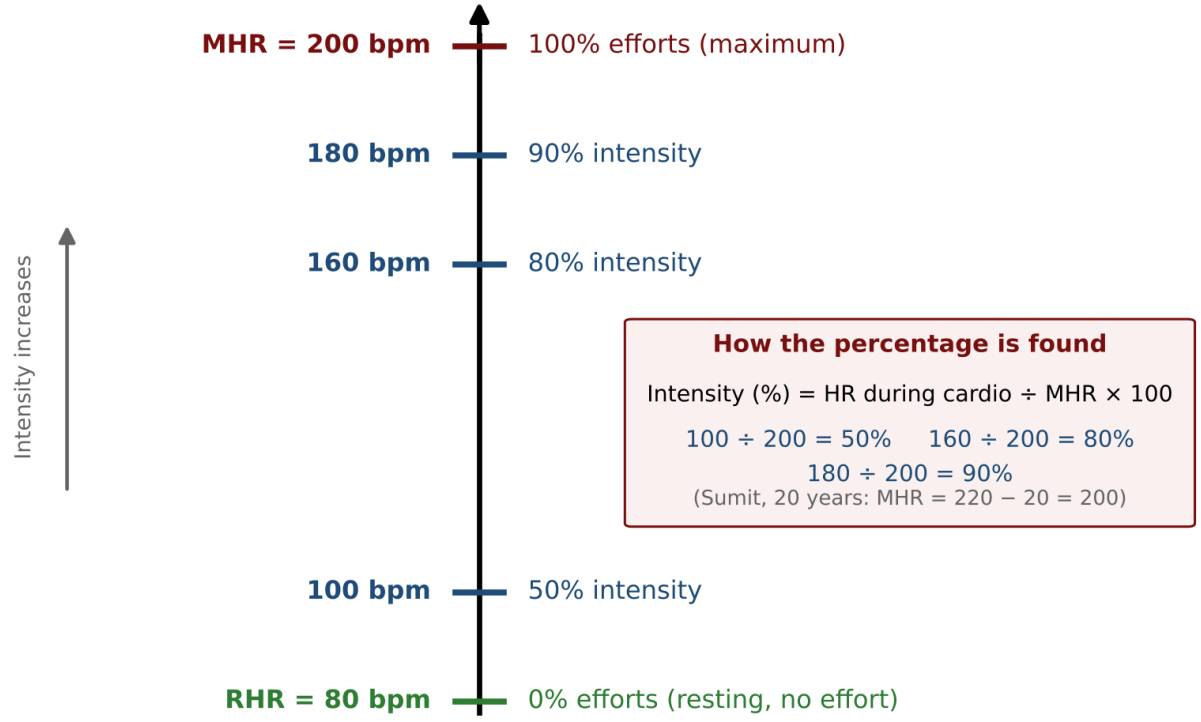
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1. Intensity in cardio

- **Workload** = the amount of work done (speed, distance, time). **Intensity** = the level of effort or difficulty of that work. Same workload, different intensity for different people (exam analogy).
- Weight training: **lower RM = higher intensity**. Cardio: **higher HR = higher intensity**.
- The HR is turned into a percentage of that person's MHR:

> **Intensity (%) = (HR during cardio ÷ MHR) × 100**

Intensity scale for the case study client (Sumit, 20 years)



The bpm positions are drawn to scale between 80 and 200. HR is divided by MHR, not by 200 in general: use each client's own MHR.

Sumit (20 years, MHR = 220 minus 20 = 200)	Intensity
HR 100	100 ÷ 200 = **50%**
HR 160	160 ÷ 200 = **80%**
HR 180	180 ÷ 200 = **90%**

- The bottom of the scale is the **RHR** (the "no effort" end) and the top is the **MHR** (100% effort).
- The numbers are only for the case study. For any client, use **that client's own MHR**.

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2. Classification and promotion

Intensity	% of MHR	Trainee level
Low	**50 to 60%**	Beginner
Moderate	**60 to 70%**	Intermediate
High	**70 to 85%**	Advanced
Very high	**Above 85%**	Super advanced

Classification of cardio intensity (percentages are of MHR)

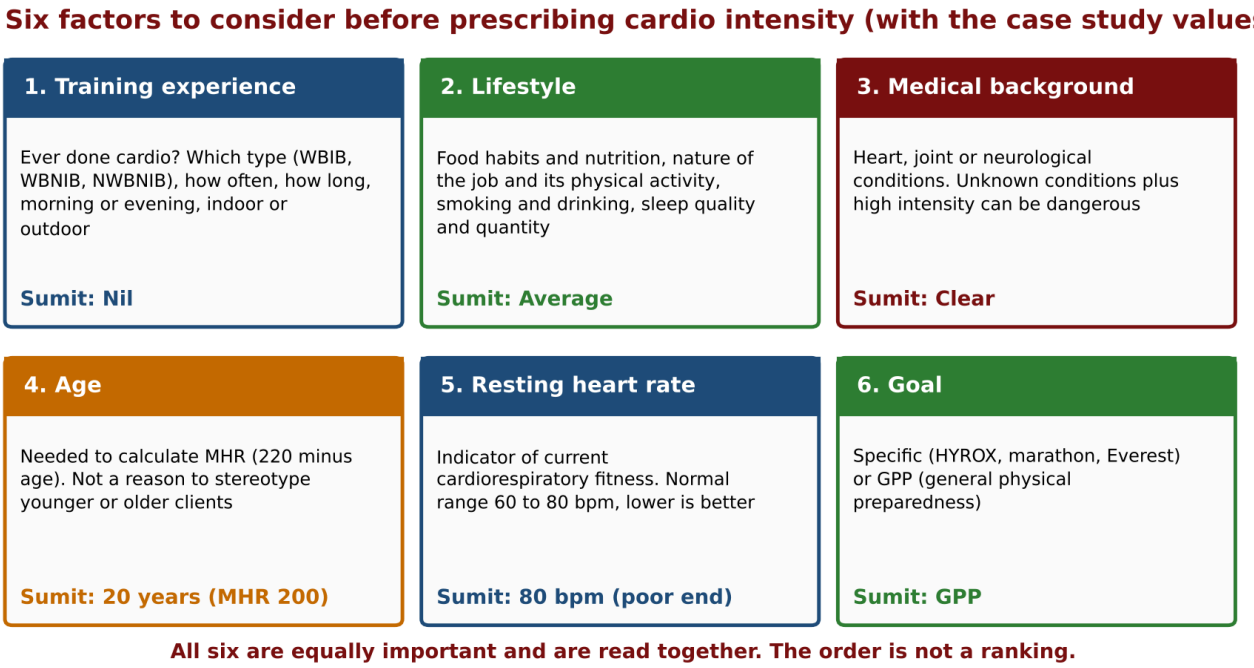
Intensity	% of MHR	Trainee level	Cardio experience (lecture guide)	Sumit's bpm*
Very high	above 85%	Super advanced	Athletes and sports people who can go above 85% safely and recover without fatigue or risk	above 170
High	70 to 85%	Advanced	After a year of consistent performance and progress, when ready to be promoted	140 to 170
Moderate	60 to 70%	Intermediate	More than 3 months but less than 1 year (promoted on performance, not the calendar)	120 to 140
Low	50 to 60%	Beginner	Less than 2 to 3 months, or restarting after a gap, illness or rehab	100 to 120

* Computed for these notes from Sumit's MHR of 200 bpm (for example 50% of 200 = 100). Not written in the lecture. Remember: do not copy these numbers for other clients.

- **Beginner**: less than about 2 to 3 months of experience. **Restarting clients are beginners** too.
- Rough guide: intermediate is more than 3 months and less than 1 year, advanced is after a year of consistent progress.
- **Promotion is by performance, not the calendar.** A fitness test on day one and after the beginner phase (Fitness Testing 1 and 2) shows whether improvement is significant.
- Super advanced = can go above 85% easily and come back safely (athletes). There is no official level above it, but "elite" (Bolt, Phelps) would be very close to MHR.
- **Safety rule:** never let a client take the ego risk of going near 100%. Step by step.
- The intensity is **not changed daily**. It is set from the client's current level and judged by consistent performance and fitness tests.

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3. Six factors for prescribing intensity



#	Factor	What it tells	Sumit
1	Training experience	Type, frequency, preferences of past cardio	Nil
2	Lifestyle	Food, job, habits, sleep (quality and quantity), stress	Average, mostly inactive
3	Medical background	Heart, joint or neurological conditions	Clear
4	Age	Needed to calculate **MHR** (not for stereotyping)	20 (MHR 200)
5	RHR	Current fitness level (normal 60 to 80, lower is better)	80
6	Goal	Specific goal, or GPP (general physical preparedness)	GPP

- All six are **equally important** and are seen **together**.
- Decision for Sumit: **beginner, 50 to 60%**.
- The client is given a **THR number**, not a percentage, because a number is easy to follow on a device.

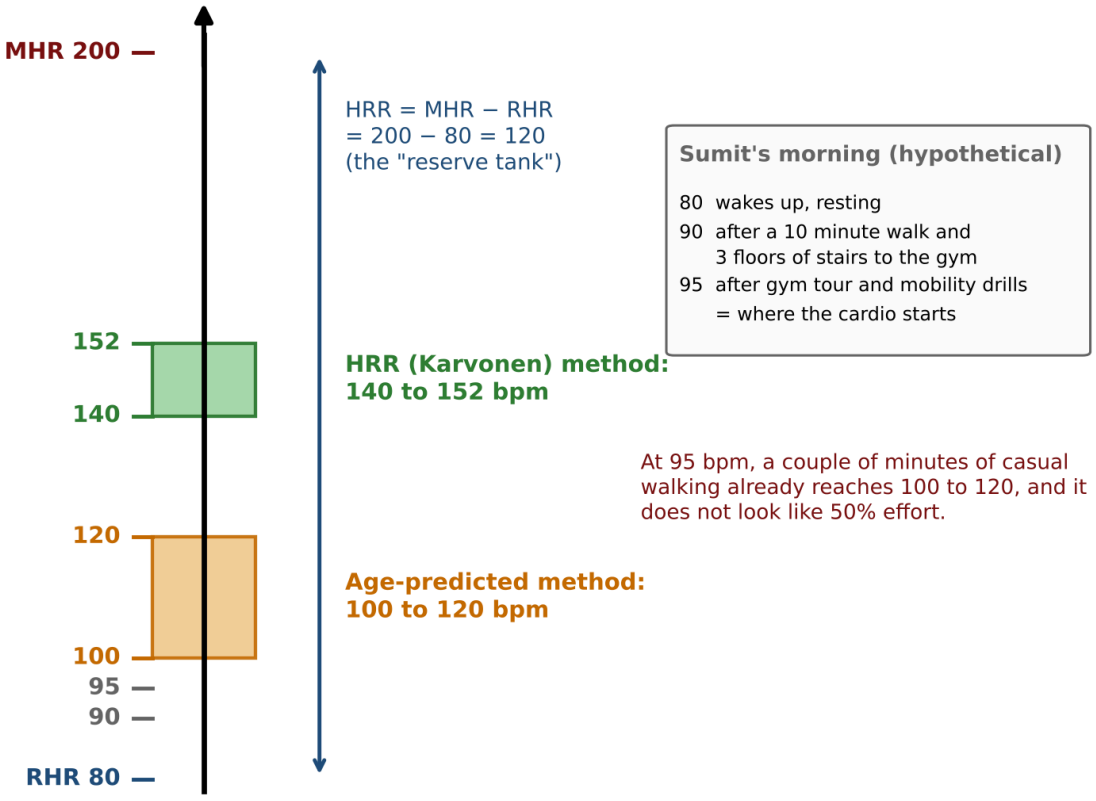
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4. Target heart rate (THR)

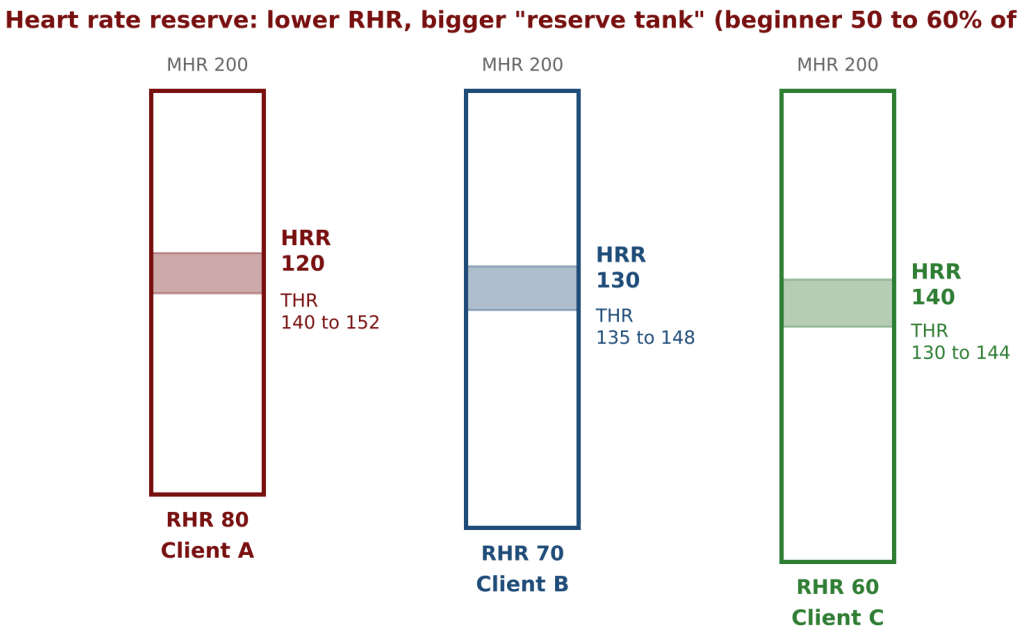
- The THR is a **range** (lower limit L and upper limit U), because heart rate fluctuates and an exact number cannot be maintained. Calculate the formula **twice**.

	Age-predicted (MHR) method	HRR (Karvonen) method
Formula	THR = (220 minus age) × intensity	THR = (220 minus age minus RHR) × intensity + RHR
Uses	Age only	Age **and** RHR
Sumit, 50%	200 × 0.5 = **100**	120 × 0.5 + 80 = **140**
Sumit, 60%	200 × 0.6 = **120**	120 × 0.6 + 80 = **152**
Range	100 to 120 bpm	**140 to 152 bpm**
Nature	Generalised	**Customised** (accepted worldwide)

Same client, two methods: where the target zone falls (Sumit, 20 years, RHR 80)



- **BODMAS:** brackets first, then multiplication, then addition. Do 120 × 50% = 60 first, then add 80.
- **The loophole:** Sumit's cardio starts at about 95 bpm (walk, stairs, drills), so 100 to 120 is reached by casual walking. All 20 year olds would get the same THR even with RHR 60, 70 or 80.



Same MHR (200), lower RHR: the reserve gets bigger (120, 130, 140). THR (shaded) is computed from each client's own RHR.
Client A has the same RHR as Sumit. The THR values for Clients B and C were computed for these notes with the same formula (not worked in the lecture).

- **HRR = MHR minus RHR** is a **reserve tank** for the heart to go up. A **bigger reserve is better**, so a **lower RHR is better**.
- MHR cannot be changed (age), but **cardio lowers the RHR** and widens the reserve.
- Guitar class story: RHR 80 means arriving exhausted and late, RHR 70 means a short wait, RHR 60 means arriving on time and fresh.
- Computed for the notes (not in the lecture): RHR 70 gives 135 to 148, RHR 60 gives 130 to 144 (age 20, 50 to 60%).

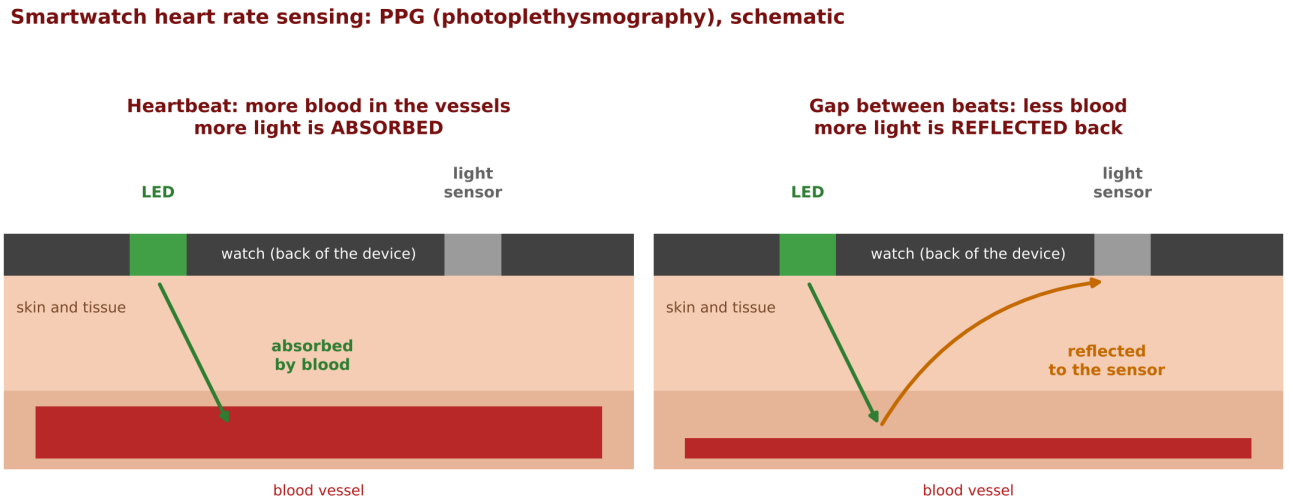
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5. Monitoring with electronic devices

Electronic devices for monitoring heart rate against the THR

	Chest strap	Smartwatch (wrist, finger, ring)	Grip sensors
Verdict	Highest accuracy	Convenient; accuracy depends on price	NOT recommended (inaccurate)
How it senses	Electrodes on the back of the main unit, just above the sternum (closest to the heart), pick up the heart's electric signals; sent by Bluetooth to a phone or watch	PPG: LEDs shine light into the skin; light sensors read how much is absorbed (blood flow high) or reflected (blood flow low). Green or red lights on the back	Sensors in the machine handles (holes under the fingers) read weak electric signals from the skin surface
Typical users	Athletes (real time, accurate data)	Gym goers	Anyone using cardio machines (common but unreliable)
Examples	Wahoo, Garmin, Polar	Apple, Samsung, Fitbit, Garmin, Xiaomi (cheaper)	Treadmill, elliptical, upright bike handles
Limits	Wearing a strap across the chest	Entry level versus high end. Basic: calories, steps, HR. Higher: VO2 max, sleep quality, GPS, indoor or outdoor modes	Weak signals; motion interference (jogging, running); salt from sweat and sanitiser residue form a barrier

Device	Strength	How	Who	Brands
Chest strap	**Most accurate**	Electrodes just above the sternum pick up the heart's electric signals, sent by Bluetooth	Athletes	Wahoo, Garmin, Polar
Smartwatch (wrist, finger, rings)	**Convenient**, accuracy varies with price	**PPG**: LEDs send light, blood absorbs more at a beat and less between beats, sensors read the reflected light	Gym goers	Apple, Samsung, Fitbit, Garmin, Xiaomi
Grip sensor (gym machines)	**Not recommended**	Sensors in the handle holes read weak skin signals	Built into machines	None



The watch compares how much light was absorbed and how much came back, and turns the pattern into a heart rate.

- Grip sensors are unreliable because of **weak signals**, **motion interference** when jogging or running, and **salt and sanitiser build-up** on the sensor.
- A basic smartwatch is an **investment in oneself**, not an overhead.

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6. Estimating intensity without devices

Without electronic devices: for whom?

Patients (heart, joint, neurological) | Pregnant women | People on beta blockers
No THR is given (risk of a wrong target), so the client self-regulates and the intensity is estimated, not calculated

I. RPE (experience based)

Rating = score, Perceived = experience,
Exertion = difficulty (Gunnar Borg, scale 0 to 10)

Client gives themselves a score and the trainer compares it with the prescribed one (for example 3).

Shortcoming: rating mismatch
- state of mind (upset, stressed)
- environment (heat, poor ventilation)

Subjective: the same speed feels different to different people

II. Talk test (observation based)

Observe two things: breathing pattern and talking

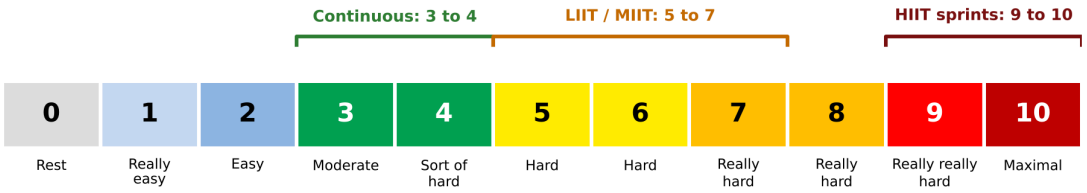
Intensity	Breathing	Talking
Low	Normally	Yes
Moderate	Deep and often	Yes
High	Deep and quick	Briefly

Helper trainer can chat with the client and watch breathing and speech while the personal trainer is busy

- For **patients, pregnant women and people on beta blockers**. Intensity is **estimated, not calculated**, and the client **self-regulates**.
- **Beta blocker**: a medicine that artificially holds the heart rate down, so a THR cannot be reached.

RPE (Gunnar Borg, 0 to 10): Rating, Perceived, Exertion. The trainer gives a rating (for example 3, moderate) and the client checks the same feeling.

Borg RPE scale (0 to 10): rating of perceived exertion



Prescribe a rating (for example "do this at 3, moderate"); the client reports the rating they feel. Ideally the two match.

Shortcoming: rating mismatch (told 3, feels 5) caused by state of mind or environment. Board grouping: 1 to 2 easy, 5 to 6 hard, 7 to 8 very hard, 9 to 10 maximal efforts.

- Rating mismatch comes from **state of mind** (stress, mood) and **environment** (heat, poor ventilation).
- It is subjective, and that is why it is called estimation.

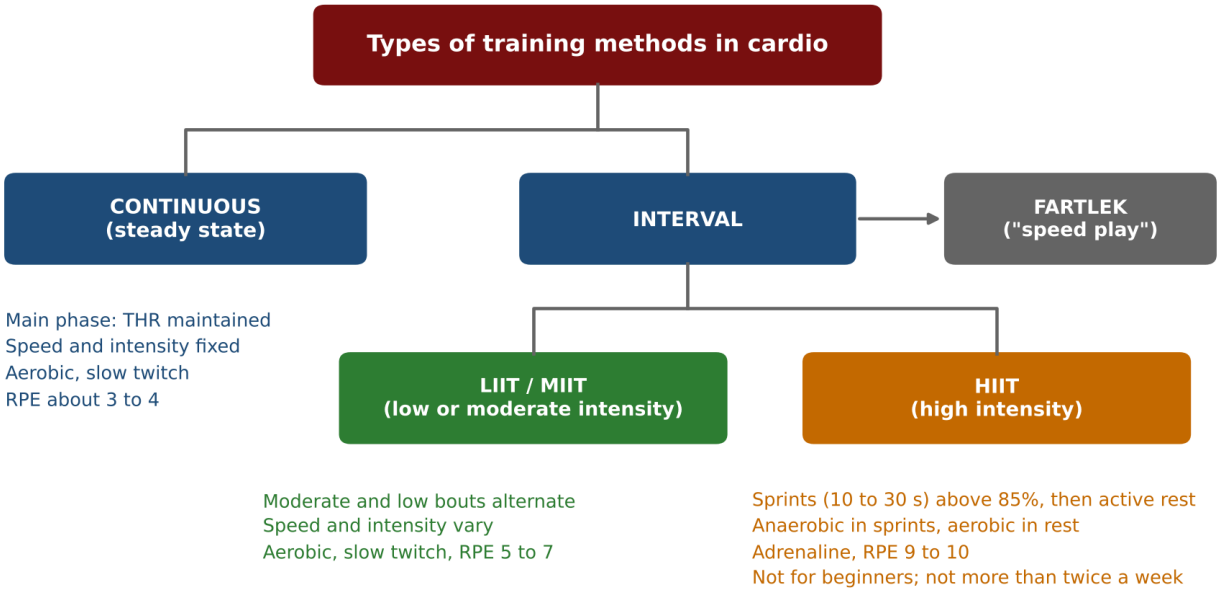
Talk test: the trainer observes breathing and talking.

Intensity	Breathing	Talking
Low	Normally	Yes
Moderate	Deep and often	Yes
High	Deep and quick	Briefly (half air, half word)

- It also lets a general trainer supervise a risk client from a distance.

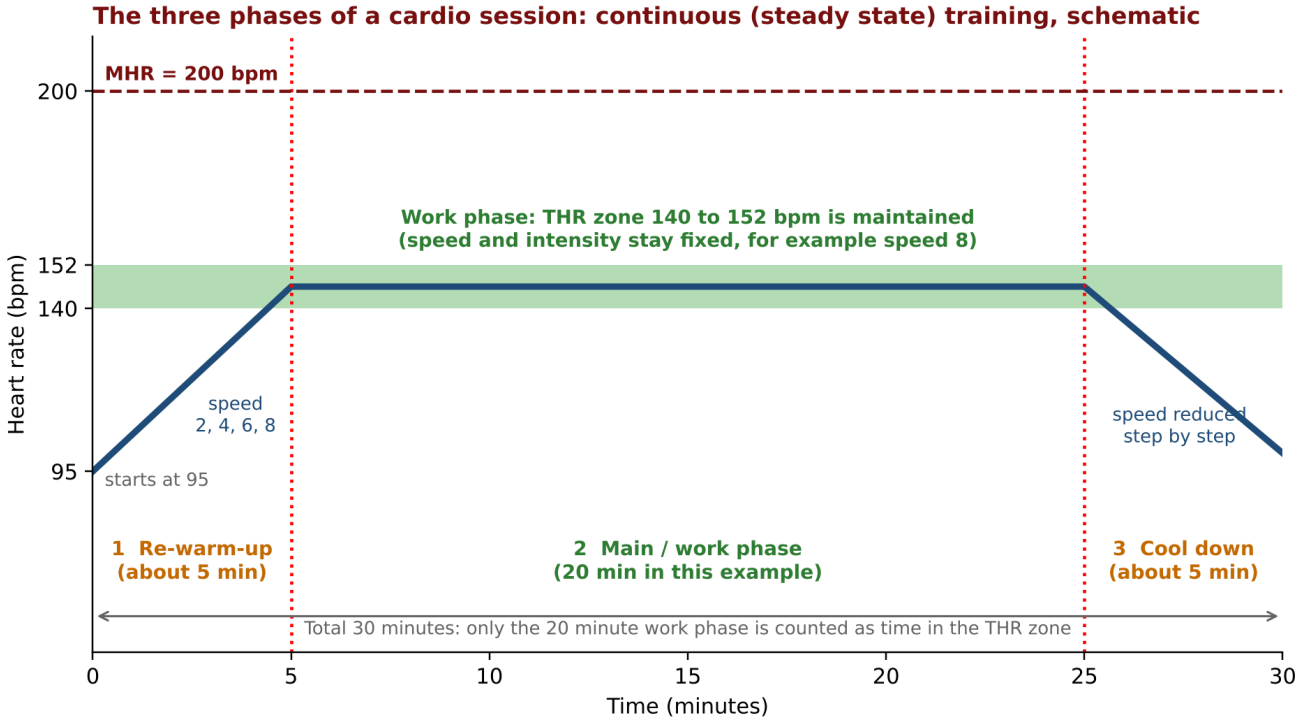
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7. Training methods

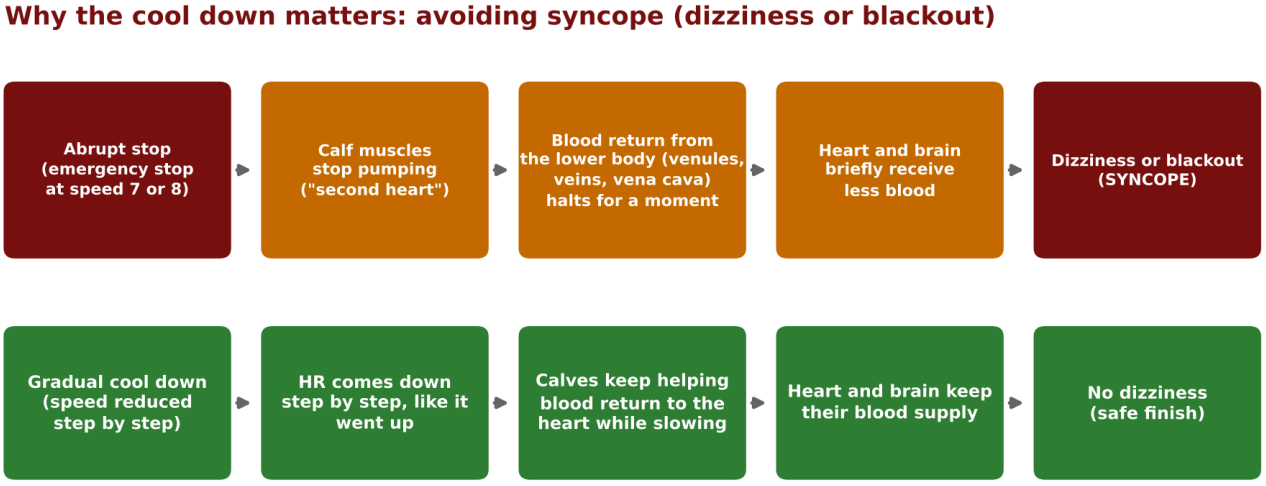


Every session has the same frame: 1 re-warm-up, 2 main (work) phase, 3 cool down. Fartlek is drawn under interval on the board.

- **Continuous (steady state)** = non-stop. **Interval** = in phases (LIIT/MIIT, HIIT, Fartlek).
- Every session has three phases: **re-warm-up** (rise, cardio specific, about 5 minutes in the example), **main phase** (20 minutes in the example), **cool down** (fall, about 5 minutes).



- **Cool down avoids syncope**: dizziness or a blackout when stopping abruptly. "Calves are the second heart": they pump blood back up, and an abrupt stop pauses that return.



	Continuous	LIIT / MIIT	HIIT
Main phase shape	Flat line	Buildings	Mountains
Speed / intensity	Fixed (THR held)	Varies (moderate and low)	Varies (sprint and active rest)
Energy pathway	Aerobic	Aerobic	Sprint anaerobic, active rest aerobic
Muscle fibres	Slow twitch	Slow twitch	Fast twitch (sprint), slow twitch (active rest)
RPE	3 to 4	5 to 7	9 to 10

Training methods compared (all share: re-warm-up, main phase, cool down)

	Continuous (steady state)	LIIT / MIIT	HIIT	Fartlek
Main phase pattern	One flat line: THR maintained	Moderate and low bouts alternate ("buildings"), e.g. jog and walk	Sprints (S) and active rests (AR) ("mountains")	Speed and intensity vary randomly, as needed (even direction)
Speed / intensity	Fixed	Varies	Varies (sprint above 85%)	Varies
Energy pathway	Aerobic	Aerobic	Sprint anaerobic, active rest aerobic	Not stated in the lecture
Muscle fibres	Slow twitch (type 1)	Slow twitch	Fast twitch in sprints, slow twitch in rest	Not stated
RPE	3 to 4	5 to 7	9 to 10 (sprints)	Not stated
Notes	Warm-up and cool-down 5 min each (example)	Beginner: jog/walk; intermediate: run/jog	Not for beginners; max twice a week; adrenaline rush	Swedish "speed play"; athletes; real-time instructions

- **LIIT/MIIT**: beginner jog (moderate) and walk (low), four jog bouts and three walk bouts. Intermediate: run (moderate) and jog (low).
- **HIIT**: sprints (S1 to S4, above 85%, 10 to 30 seconds, adrenaline) with **active rest** (AR1 to AR3) at moderate intensity, such as jogging (not walking, not running). Sprint to rest ratio **1:2 or 1:3**. **Not for beginners**, **not more than twice a week**.
- **Fartlek** ("speed play", Sweden): random changes in speed and direction, used for sports. The trainer calls the pace using a chest strap reading.
- **Six ways to progress:** speed, distance, duration, angle, resistance level, gradient.

Six ways to progress in cardio

Factors of workload (from Cardio Regulations 1)

1. Speed	2. Distance	3. Duration (time)
increase the speed (for example treadmill 6.0 to 8.0)	cover a longer distance	exercise for longer (for example 15 to 20 min)

Other ways (good to know)

4. Angle	5. Resistance level	6. Gradient
change from flat to incline (treadmill)	turn the knob (bike, cross trainer)	run uphill outdoors instead of flat

8. Key terms

Term	Meaning
Intensity (cardio)	HR during cardio as a percentage of MHR
MHR	220 minus age
RHR	Heart rate at complete rest (normal 60 to 80)
HRR	MHR minus RHR, the heart's reserve
THR	Target heart rate range (L and U)
Age-predicted (MHR) method	THR = (220 minus age) × intensity
HRR / Karvonen method	THR = (220 minus age minus RHR) × intensity + RHR
GPP	General physical preparedness (basic fitness goal)
PPG	Light-based heart rate sensing in smartwatches
RPE	Rating of perceived exertion, Gunnar Borg 0 to 10
Talk test	Judging intensity by breathing and talking pattern
Re-warm-up	Cardio-specific warm-up that raises HR to the zone
Syncope	Dizziness or blackout after an abrupt stop
LIIT / MIIT	Low or moderate intensity interval training
HIIT	High intensity interval training (sprints and active rest)
Active rest	Moderate activity between sprints, not full rest
Fartlek	"Speed play": random changes in speed and direction

9. Where the sources differ

- **Beginner band and the HRR method:** 50 to 60% is applied to HRR, giving 140 to 152 bpm for Sumit, which is 70 to 76% of MHR (the high band of the classification). The lecture does not reconcile this.
- **The 85% boundary:** the lecture says high is 70 to 85% and very high is above 85%. The slide writes "greater than or equal to 85%" for very high.
- **HIIT versus classification:** sprints go above 85% (super advanced band), yet the lecture allows intermediate and advanced clients to do HIIT (twice a week at most).
- **Time guide for levels:** the lecture gives months. The slides give none. The E-book gives weeks for its phases.
- **Graph timing:** the HIIT graph is schematic (four sprints and three rests would not fill 20 minutes). LIIT/MIIT bouts of 2 to 3 minutes do not exactly add up to 20.
- **Fartlek** is under interval in the lecture, but the E-book lists it as a separate modality.
- **RPE labels** differ slightly between the lecture grouping and the slide wording.
- The THR values for the clients with RHR 70 and 60 were computed for the notes, not taken from the lecture. See the Detailed Notes, Points to Confirm, for the full list.

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10. Quick revision questions

1. What is the difference between workload and intensity in cardio? Give the exam analogy.
2. What deciding value is used for intensity in cardio, and how does it compare with weight training?
3. A client has an MHR of 200. Find the intensity at HR 100, 160 and 180.
4. State the four cardio intensity levels, their percentage ranges and the matching trainee levels.
5. Why is a client not promoted just because three months have passed? What is used instead?
6. List the six factors used to prescribe cardio intensity.
7. Why is age needed as a factor? What stereotype should be avoided?
8. Why is the THR given as a range and not as a single number?
9. Write the age-predicted (MHR) method formula and calculate the THR for a 20 year old beginner.
10. What is the loophole in the age-predicted method?
11. Write the HRR (Karvonen) formula and calculate the THR for a 20 year old beginner with an RHR of 80.
12. Why must the brackets and multiplication be done before the addition in the Karvonen formula?
13. Why is a lower RHR better? Which part of HRR can be changed, and how?
14. A client is 30 years old with an RHR of 70 and is an intermediate (60 to 70%). Find the THR with both methods.
15. Compare the chest strap, smartwatch and grip sensor on accuracy, who uses them and how they sense the heart rate.
16. Give three reasons grip sensors are not reliable.
17. Who uses estimation methods instead of a THR, and why?

18. What is RPE, and what two factors can cause a mismatch between the trainer's rating and the client's?
19. Describe the talk test for low, moderate and high intensity.
20. Compare continuous, LIIT/MIIT and HIIT (main phase shape, energy pathway, fibres, RPE), and state the HIIT rules. Name the six ways to progress.

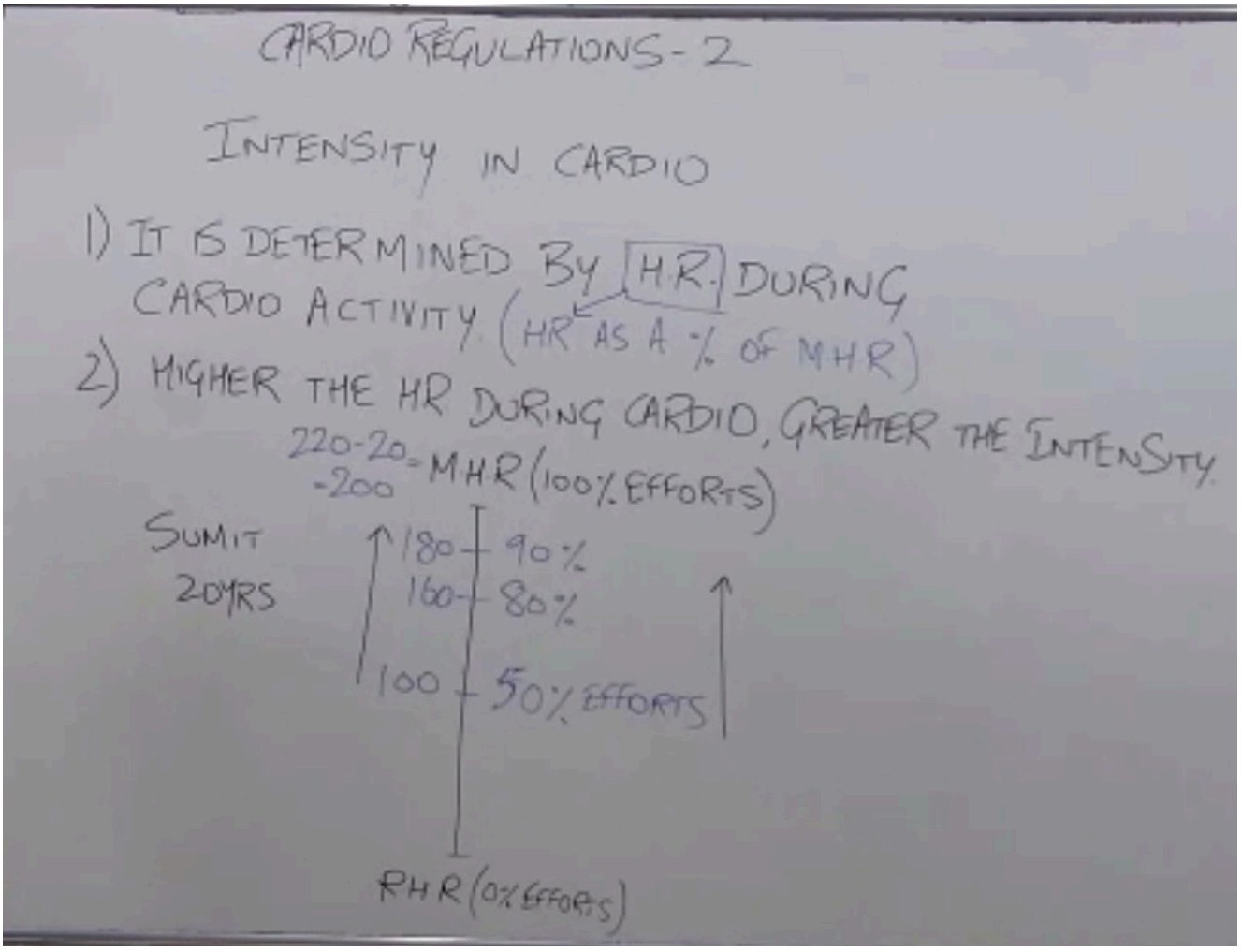
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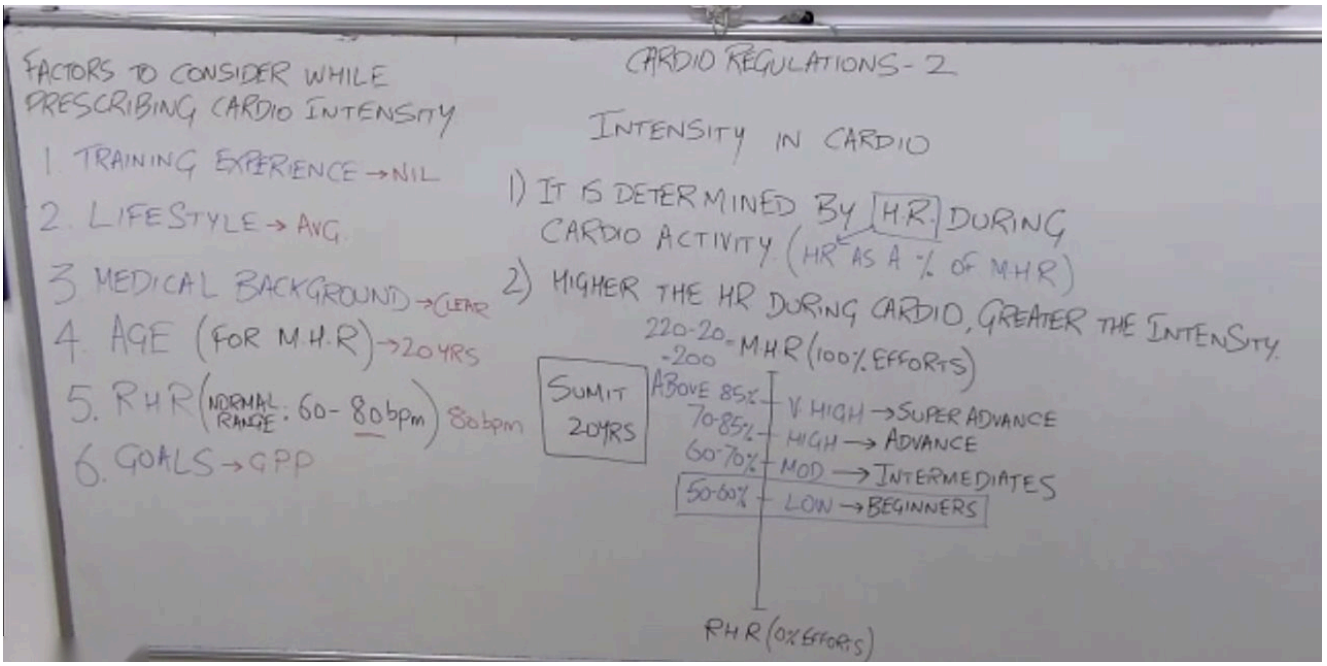
11. Answers

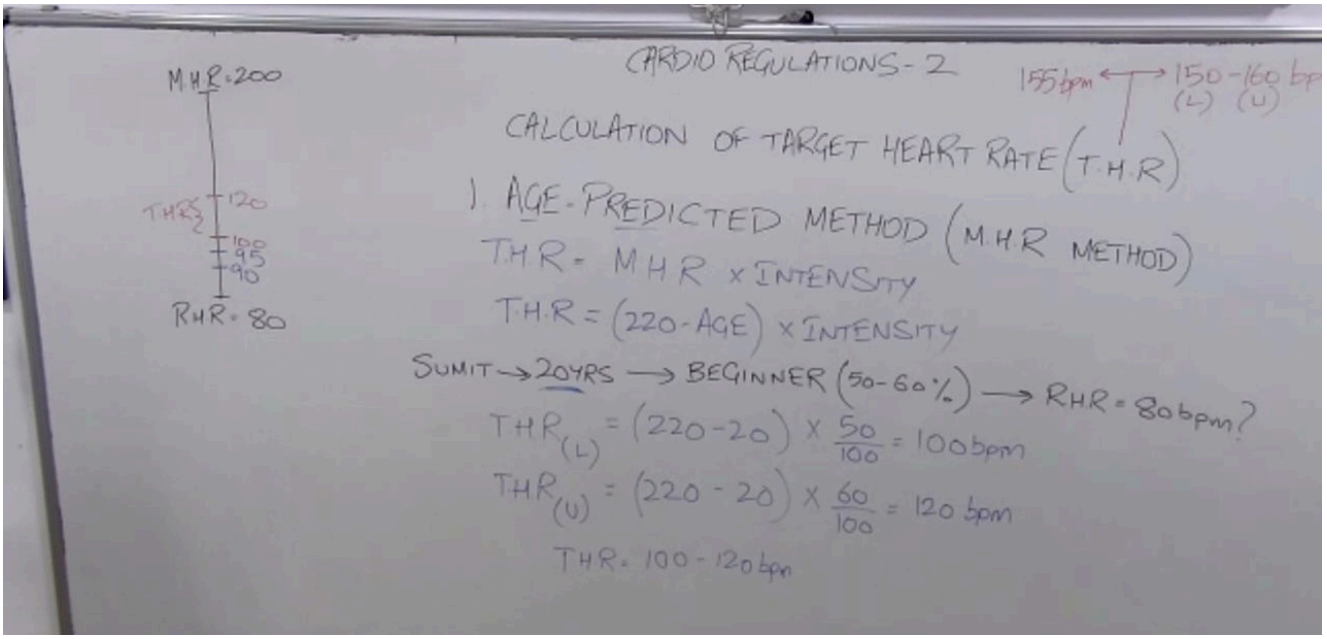
1. **Workload** is the amount of work done (speed, distance, time). **Intensity** is the level of effort or difficulty. In the exam analogy, 30 students solve the same 30 questions (same workload), but a well-prepared student puts in low effort (low intensity) and an unprepared one puts in high effort (high intensity).
2. The **heart rate during the cardio activity, as a percentage of MHR**. In weight training a **lower RM means higher intensity**. In cardio a **higher HR means higher intensity**.
3. Intensity = HR ÷ MHR × 100: 100 ÷ 200 = **50%**, 160 ÷ 200 = **80%**, 180 ÷ 200 = **90%**.
4. **Low 50 to 60% (beginner), moderate 60 to 70% (intermediate), high 70 to 85% (advanced), very high above 85% (super advanced).**
5. Time alone does not show progress. A client may be irregular, inattentive or not following the plan in the other 23 hours. A **fitness test** on day one and after the beginner phase (Fitness Testing 1 and 2) shows significant improvement, and that is when the client is promoted.
6. **Training experience, lifestyle, medical background, age, resting heart rate, goal.** All six are equally important.
7. Age is needed to calculate **MHR (220 minus age)**, and without MHR intensity cannot be found. It should not be used to stereotype (for example assuming every 60 year old is a beginner and every 20 year old is fit).
8. Heart rate **fluctuates** by a few beats, so an exact number cannot be maintained. A range (lower and upper limit) tells the client to stay within it. (Shirt price analogy.)
9. THR = (220 minus age) × intensity. MHR = 200. Lower: 200 × 50/100 = **100**. Upper: 200 × 60/100 = **120**. Range **100 to 120 bpm**.
10. It uses **only age**, ignoring the client's current fitness level (RHR). Sumit's heart rate is already about 95 before cardio, so 100 to 120 is reached by casual walking, and every 20 year old gets the same THR whatever the RHR (jeans analogy).
11. THR = (220 minus age minus RHR) × intensity + RHR. HRR = 220 minus 20 minus 80 = 120. Lower: 120 × 0.5 + 80 = 60 + 80 = **140**. Upper: 120 × 0.6 + 80 = 72 + 80 = **152**. Range **140 to 152 bpm**.
12. BODMAS: brackets first, then multiplication, then addition. Adding 80 to 120 before multiplying would give a completely different (wrong) value. Right order: 120 × 0.5 = 60, then 60 + 80 = 140.
13. HRR = MHR minus RHR is the heart's reserve tank, so a lower RHR means a bigger reserve and the same work feels easier and leaves the client further from exhaustion. MHR depends on age and cannot be changed, but **cardio lowers the RHR**, which widens the HRR.
14. MHR = 220 minus 30 = 190. **Age-predicted:** 190 × 0.6 = 114 and 190 × 0.7 = 133, so **114 to 133 bpm**. **Karvonen:** HRR = 190 minus 70 = 120. Lower: 120 × 0.6 + 70 = 72 + 70 = 142. Upper: 120 × 0.7 + 70 = 84 + 70 = 154. So **142 to 154 bpm**.

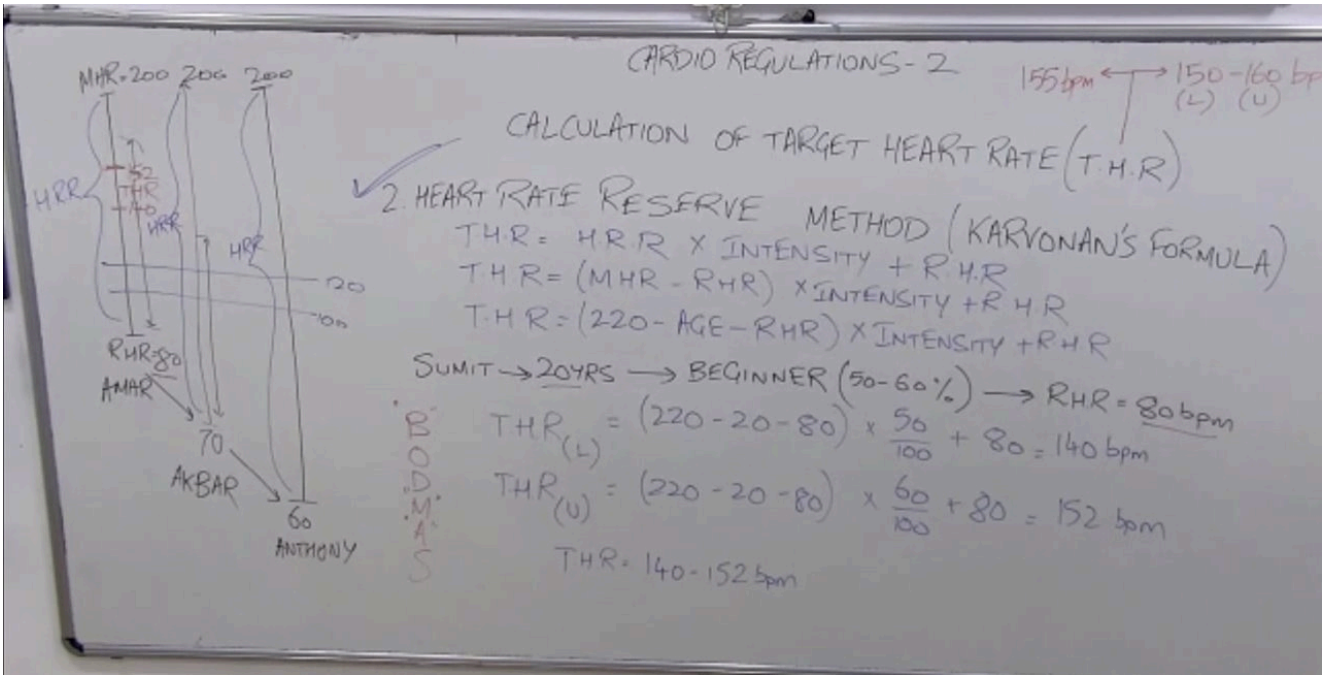
15. **Chest strap:** highest accuracy, electrodes just above the sternum read the heart's electric signals and send them by Bluetooth, used by athletes. **Smartwatch:** convenient, accuracy depends on price, uses PPG (LED light absorbed and reflected by blood, read by light sensors), used by gym goers. **Grip sensor:** built into machines, reads weak skin signals through handle holes, not recommended.
16. **Weak signals** at the skin surface, **motion interference** when jogging or running, and **salt from sweat and sanitiser residue** forming a barrier on the sensor.
17. **Patients, pregnant women and people on beta blockers** (the slides add cardiac and diabetic patients and those without monitors). A THR could push them into risk, or a beta blocker holds the heart rate down so the THR is never reached. They **self-regulate** using RPE or the talk test.
18. **RPE** = Rating of Perceived Exertion (Gunnar Borg, 0 to 10): the client rates how hard the activity feels. A mismatch (for example trainer says 3, client says 5) comes from **state of mind** (such as upsetting news) and **environment** (such as heat or poor ventilation).
19. **Low:** breathing normally, can talk. **Moderate:** deep and often, can still talk. **High:** deep and quick, can talk only briefly (half air, half word).
20. **Continuous:** flat line, aerobic, slow twitch, RPE 3 to 4, THR held. **LIIT/MIIT:** buildings (moderate and low), aerobic, slow twitch, RPE 5 to 7. **HIIT:** mountains (sprints above 85% and active rest), sprint anaerobic with fast twitch, active rest aerobic with slow twitch, RPE 9 to 10. HIIT rules: not for beginners, not more than twice a week, sprint to rest ratio 1:2 or 1:3, active rest at moderate intensity (jog). Six ways to progress: **speed, distance, duration, angle, resistance level, gradient**.

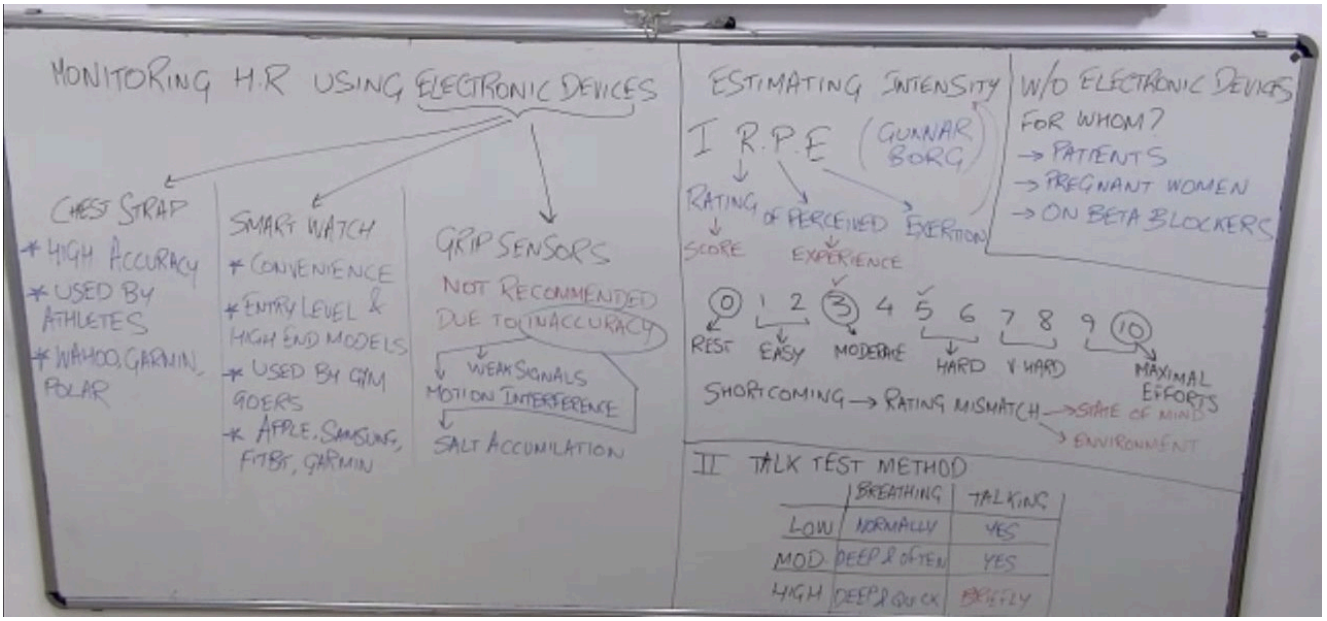
Board Photos from the Lecture

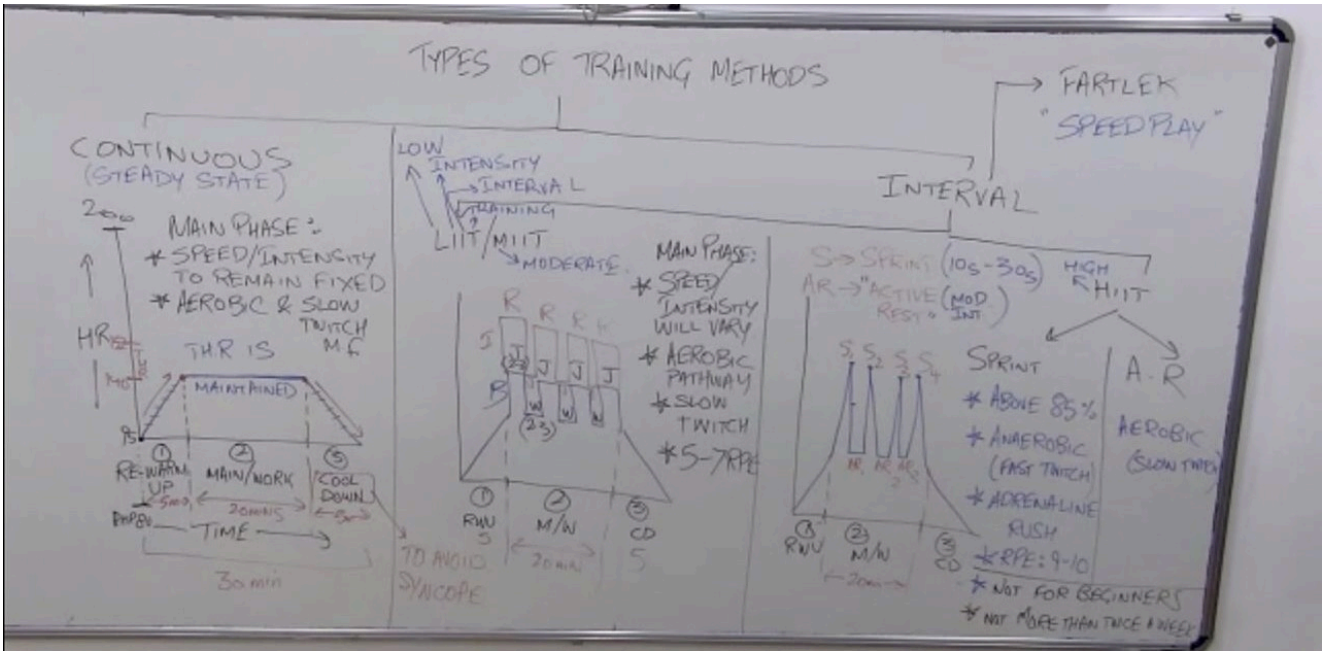





CARDIO REGULATIONS - 2
155 bpm
150 - 160 bpm
(L) (U)
CALCULATION OF TARGET HEART RATE (T.H.R)
1. AGE-PREDICTED METHOD (M.H.R METHOD)
T.H.R = M.H.R x INTENSITY
T.H.R = (220 - AGE) x INTENSITY
SUMIT → 20 YRS → BEGINNER (50-60%) → RHR = 80 bpm?
THR(L) = (220 - 20) x 50/100 = 100 bpm
THR(U) = (220 - 20) x 60/100 = 120 bpm
THR = 100 - 120 bpm
MHR = 200
THR
120
100
95
90
RHR = 80


CARDIO REGULATIONS - 2
155 bpm
150 - 160 bpm
(L) (U)
CALCULATION OF TARGET HEART RATE (T.H.R)
2. HEART RATE RESERVE METHOD (KARVONAN'S FORMULA)
T.H.R = H.R.R x INTENSITY + R.H.R
T.H.R = (MHR - RHR) x INTENSITY + R.H.R
T.H.R = (220 - AGE - RHR) x INTENSITY + RHR
SUMIT → 20 YRS → BEGINNER (50-60%) → RHR = 80 bpm
THR(L) = (220 - 20 - 80) x 50/100 + 80 = 140 bpm
THR(U) = (220 - 20 - 80) x 60/100 + 80 = 152 bpm
THR = 140 - 152 bpm
MHR = 200
HRR
THR
RHR = 80
AMAR
70
AKBAR
60
ANTHONY
BODMAS


MONITORING H.R USING ELECTRONIC DEVICES
CHEST STRAP
* HIGH ACCURACY
* USED BY ATHLETES
* WAHOO, GARMIN, POLAR
SMART WATCH
* CONVENIENCE
* ENTRY LEVEL & HIGH END MODELS
* USED BY GYM GOERS
* APPLE, SAMSUNG, FITBIT, GARMIN
GRIP SENSORS
NOT RECOMMENDED DUE TO INACCURACY
WEAK SIGNALS
MOTION INTERFERENCE
SALT ACCUMULATION
ESTIMATING INTENSITY
I R.P.E (GUNNAR BORG)
RATING OF PERCEIVED EXERTION
SCORE
EXPERIENCE
0 1 2 3 4 5 6 7 8 9 10
REST
EASY
MODERATE
HARD
V HARD
MAXIMAL EFFORTS
SHORTCOMING → RATING MISMATCH → STATE OF MIND
→ ENVIRONMENT
II TALK TEST METHOD
BREATHING
TALKING
LOW NORMALLY YES
MOD DEEP & OFTEN YES
HIGH DEEP & QUICK BRIEFLY
W/O ELECTRONIC DEVICES
FOR WHOM?
→ PATIENTS
→ PREGNANT WOMEN
→ ON BETA BLOCKERS


TYPES OF TRAINING METHODS
CONTINUOUS (STEADY STATE)
MAIN PHASE:
* SPEED/INTENSITY TO REMAIN FIXED
* AEROBIC & SLOW TWITCH M.F
THR IS MAINTAINED
RE-WARM UP
MAIN/WORK
COOL DOWN
TIME
30 min
LOW INTENSITY INTERVAL TRAINING
LIIT/MIIT
MODERATE
TO ANGIO SYNCOPE
MAIN PHASE:
* SPEED/INTENSITY WILL VARY
* AEROBIC PATHWAY
* SLOW TWITCH
* 5-7 RPE
INTERVAL
FARTLEK "SPEED PLAY"
S → SPRINT (10s-30s)
AR → "ACTIVE REST" (MOD INT)
HIGH INT HIIT
SPRINT
* ABOVE 85%
* ANAEROBIC (FAST TWITCH)
* ADRENALINE RUSH
* RPE: 9-10
* NOT FOR BEGINNERS
* NOT MORE THAN TWICE A WEEK
A.R
AEROBIC (SLOW TWITCH)